

PROFESSIONAL FLASHCARDS (1–10)

■ FLASHCARD 1

What is Generative AI (Gen AI)?

Answer:

Generative AI is a class of artificial intelligence that enables machines to **generate original content** such as text, images, audio, video, and code by learning patterns from large datasets.

Unlike traditional AI systems that focus on classification or prediction, Generative AI models are designed to **produce new data that resembles the training data**.

Key Insight:

Generative AI shifts AI from *analysis* → *creation*.

Examples:

- AI chatbots generating responses
 - Image generation from text prompts
 - Automated code generation
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■ FLASHCARD 2

How does Generative AI work?

Answer:

Generative AI operates using deep learning models trained on large-scale datasets.

Workflow:

1. **Training Phase**
The model learns patterns, grammar, structure, and relationships in data.
2. **Prompt Input**
The user provides an input (text, image, etc.).
3. **Processing**
The model interprets the prompt using learned representations.
4. **Content Generation**
It predicts and generates the most relevant output.

Core Technologies:

- Neural Networks
 - Transformer Architecture
 - Token-based prediction
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■ FLASHCARD 3

What are the key components of Generative AI?

Answer:

Component	Description
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Model	The AI system (e.g., LLMs) responsible for generation
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Training Data	Large datasets used to train the model
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Prompt	Input provided by the user
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Output	Generated content
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Algorithms	Mathematical methods used for learning
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Key Insight:

The quality of output depends heavily on **data quality + prompt clarity**.

■ FLASHCARD 4

What are the applications of Generative AI?

Answer:

Generative AI is transforming multiple domains:

- **Software Development** → Code generation, debugging
- **Content Creation** → Blogs, reports, scripts
- **Design & Media** → Image and video generation
- **Education** → Personalized tutoring
- **Business Automation** → Report generation, email drafting

Industry Impact:

It significantly improves productivity and reduces manual effort.

■ FLASHCARD 5

What are the advantages and limitations of Generative AI?

Answer:

Advantages:

- Enhances productivity and creativity
- Automates repetitive tasks
- Scalable and efficient
- Supports rapid prototyping

Limitations:

- Can produce incorrect or misleading outputs (hallucination)
- Bias due to training data
- High computational requirements
- Ethical and security concerns

Key Insight:

Human validation is still essential.

■ FLASHCARD 6

What is Agentic AI?

Answer:

Agentic AI refers to AI systems that can **autonomously plan, make decisions, and execute tasks** to achieve a defined goal.

Unlike Generative AI, which is reactive, Agentic AI is **proactive and goal-driven**.

Core Characteristics:

- Autonomy
 - Decision-making
 - Task execution
 - Iterative improvement
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■ FLASHCARD 7

How does Agentic AI work?

Answer:

Agentic AI follows a continuous decision-making loop:

Execution Cycle:

1. **Goal Definition**
2. **Planning**
3. **Action Execution**
4. **Observation**
5. **Iteration & Improvement**

Key Insight:

This loop enables AI systems to behave like intelligent agents capable of handling complex workflows.

FLASHCARD 8

What are the core components of an Agentic AI system?

Answer:

Component Role

LLM (Brain) Decision-making and reasoning

Memory Stores past interactions and context

Planner Breaks goals into steps

Executor Performs actions

Tools/APIs External integrations for real-world tasks

Key Insight:

The integration of these components enables **autonomous task completion**.

FLASHCARD 9

What is Notion?

Answer:

Notion is an all-in-one productivity platform that combines note-taking, task management, databases, and collaboration tools into a unified workspace.

Core Purpose:

To centralize information and workflows in a single system.

Use Cases:

- Documentation
 - Project management
 - Knowledge base creation
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 FLASHCARD 10**What are the key features of Notion?****Answer:**

- **Note-taking & Documentation** → Structured content creation
- **Databases** → Tables, Kanban boards, calendars
- **Task Management** → Track and organize work
- **Collaboration** → Real-time team editing
- **Page Linking** → Interconnected knowledge system

Key Insight:

Notion acts as a **flexible digital workspace** adaptable to personal and professional needs.